The odd-degree derivative-root mean identity

A solution to MathDB 336466

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Abstract

We prove the conjectured alternating binomial relation between the means of an odd-degree polynomial evaluated at the roots of its successive derivatives. The proof interpolates the means as a polynomial in the degree of the derivative. Its otherwise missing endpoint is an alternating coefficient pairing, which vanishes in odd degree. The argument includes arbitrary complex coefficients and roots counted with multiplicity, and gives an explicit correction in every even degree.

Task 1: the original problem and its full solution

Let $f \in \mathbb{C}[x]$ have degree $D \geq 1$. For $0 \leq \rho < D$, define

$$M_\rho(f) = \frac{1}{D-\rho} \sum_{f^{(\rho)}(r)=0} f(r),$$

where roots are counted with their algebraic multiplicities. In the notation of Wojnar, Wojnar and Brin [1], this is $\varphi_{D,0,\rho}$. Their Conjecture 2 (page 9) asserts that

$$\sum_{\rho=1}^{D-1} (-1)^\rho \binom{D}{\rho} M_\rho(f) = 0 \qquad (D \text{ odd}). \tag{1}$$

We prove the following stronger formula.

Theorem 1. *Write $f(x) = ag(x)$, where $a \neq 0$ and $g(x) = \sum_{j=0}^{D} c_j x^{D-j}$ is monic, so $c_0 = 1$. Then*

$$\sum_{\rho=1}^{D-1} (-1)^\rho \binom{D}{\rho} M_\rho(f) = a(-1)^{D+1} J_D(c), \qquad J_D(c) = \sum_{m=0}^{D} \frac{(-1)^m c_m c_{D-m}}{\binom{D}{m}}. \tag{2}$$

In particular, (1) *holds for every odd D.*

Polynomial interpolation of root power sums

For an indeterminate u and $0 \leq j \leq D$, set

$$A_j(u) = c_j \frac{(u)_{\underline{j}}}{(D)_{\underline{j}}}, \qquad (u)_{\underline{j}} = u(u-1)\cdots(u-j+1), \qquad (u)_{\underline{0}} = 1.$$

For each integer $1 \le n \le D$, direct differentiation gives

$$g_n(x) := \frac{n!}{D!} g^{(D-n)}(x) = \sum_{j=0}^{n} A_j(n) x^{n-j}. \tag{3}$$

This is monic of degree n, and $A_j(n) = 0$ for $j > n$. Introduce the formal power series

$$B(u,z) = \sum_{j=0}^{D} A_j(u) z^j, \qquad P_m(u) = -m[z^m] \log B(u,z) \quad (1 \le m \le D).$$

Here $A_0 = 1$, so the formal logarithm is well-defined. All coefficient extractions involve only finite sums; analytic convergence is irrelevant.

Lemma 1. *Each P_m is a polynomial in u of degree at most m, divisible by u. If $r_1, \dots, r_n$ are the roots of g_n, with multiplicities, then $P_m(n) = \sum_{i=1}^{n} r_i^m$.*

Proof. For integer $n \ge 1$, (3) implies $B(n,z) = \prod_{i=1}^{n}(1 - r_i z)$. Taking formal logarithms and using $\log(1 - rz) = -\sum_{m \ge 1} r^m z^m / m$ proves the power-sum identity, including repeated roots.

In the coefficient of z^m in $\log(1 + \sum_{j \ge 1} A_j(u) z^j)$, each product has indices whose sum is m. Since $\deg_u A_j \le j$, its degree is at most m. Finally, $A_j(0) = 0$ for $j \ge 1$, so $B(0,z) = 1$ and $P_m(0) = 0$. □

Consequently,

$$H(u) = c_D + \sum_{m=1}^{D} c_{D-m} \frac{P_m(u)}{u} \tag{4}$$

is a polynomial of degree at most $D - 1$. The quotients in this definition are polynomial quotients, so evaluation at $u = 0$ is legitimate. For $1 \le n \le D$, the lemma gives

$$H(n) = \frac{1}{n} \sum_{i=1}^{n} g(r_i) = M_{D-n}(g). \tag{5}$$

In particular, $H(D) = M_0(g) = 0$.

The missing endpoint and the finite difference

At $u = 0$, differentiating the formal logarithm yields

$$P_m'(0) = -m[z^m] \left. \frac{\partial B / \partial u}{B} \right|_{u=0} = -m A_m'(0).$$

The derivative of $(u)_{\underline{m}}$ at zero is $(-1)^{m-1}(m-1)!$. It follows that

$$P_m'(0) = \frac{(-1)^m m! c_m}{(D)_{\underline{m}}} = \frac{(-1)^m c_m}{\binom{D}{m}}.$$

Substitution into (4) proves

$$H(0) = c_D + \sum_{m=1}^{D} c_{D-m} P_m'(0) = J_D(c). \tag{6}$$

Proof of Theorem 1. The Dth finite difference of a polynomial of degree at most $D-1$ is zero. Apply this to H:

$$0 = \sum_{\rho=0}^{D} (-1)^\rho \binom{D}{\rho} H(D-\rho) = \sum_{\rho=1}^{D-1} (-1)^\rho \binom{D}{\rho} M_\rho(g) + (-1)^D J_D(c).$$

We used $H(D) = 0$, (5), and (6). Since multiplying by $a \neq 0$ leaves derivative roots unchanged and multiplies every mean by a, this proves (2).

If D is odd, the terms with indices m and $D-m$ in J_D have equal denominators and opposite signs. There is no unpaired middle index, so $J_D = 0$. This proves (1). For $D = 1$ the asserted sum is empty and the same proof applies. □

For example, the cubic relation is $-3M_1 + 3M_2 = 0$, and the quintic relation is $-5M_1 + 10M_2 - 10M_3 + 5M_4 = 0$. In degree two, (2) reads $-2M_1(f) = a(c_1^2/2 - 2c_2)$, confirming the sign of the even-degree correction. No assumption of distinct or real roots was used.

Task 2: the dimension of all universal mean relations

For each D, classify the vectors $(b_1, \ldots, b_{D-1})$ for which $\sum b_\rho M_\rho(f) = 0$ for every degree-D polynomial. The source's computations suggest dimension one for even $D > 2$ and dimension two for odd $D > 3$. Prove or correct these dimension claims, with all small-degree exceptions stated explicitly.

Task 3: explicit bases for the relation spaces

Find closed formulas for a basis of each space in Task 2. In odd degree, complete the binomial vector proved here to a basis. Determine whether the coefficients of $H(u)$ provide a uniform construction, and recover the source's tabulated low-degree relations as checks.

Task 4: other polynomials evaluated at derivative roots

Fix D and $0 \leq s \leq D$. Replace the value $f(r)$ by an arbitrary polynomial $h(r)$ of degree at most s, while retaining the roots of $f^{(D-n)}$. Determine the universal finite-difference identities for these means, including their polynomial continuation at $n = 0$, and characterize the choices of h for which that endpoint vanishes for every f.

Task 5: relations involving antiderivatives

The source also studies means over roots of antiderivatives and proves independence from their integration constants for the relevant polynomial evaluations. Building on that result, classify all universal linear relations for a fixed finite collection of derivative and antiderivative orders. Determine whether one interpolating polynomial accounts for the whole collection.

Verification and status. The accompanying exact verifier checks generic symbolic coefficients through degree nine and separately computes means using traces of polynomial evaluations on companion matrices. The tests include even degrees, nonmonic polynomials, complex coefficients, and repeated roots. They supplement the all-degree proof. Tasks 2–5 are proposed directions and receive no solved credit. The complete 27-page current primary and a bounded prior-claim search

were checked by the authoring agent; no independent peer review or proof assistant verification is claimed.

References

[1] G. G. Wojnar, D. Sz. Wojnar and L. Q. Brin, *Universal Peculiar Linear Mean Relationships in All Polynomials*, arXiv:1706.08381v2 (23 October 2017), definitions on pages 1–2 and Conjecture 2 on page 9. The PDF title omits "Peculiar." `https://arxiv.org/abs/1706.08381`.